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Comment on HHS-ONC-2026-0001

Governing Clinical Artificial Intelligence as a Population-Level

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Clinical artificial intelligence (AI) is no longer a discrete technology to be evaluated solely at deployment, but a population-level intervention whose safety, effectiveness, and equity can only be governed through continuous, post-deployment evaluation and accountability. I appreciate the opportunity to comment on HHS-ONC-2026-0001 and submit this response as an epidemiologist with expertise in artificial intelligence, causal inference, and population health, informed by direct experience evaluating AI systems after they have been deployed in clinical and public-sector environments. My intent is not to argue for or against particular technologies, but to address a foundational regulatory issue that underlies many of the concerns raised in this Request for Information: the growing mismatch between how clinical AI is currently regulated and how it actually behaves once implemented at scale. This comment responds directly to the Department's focus on improving the evaluation, governance, and patient-centered use of AI in health and public health, with particular attention to how AI systems should be assessed beyond pre-deployment testing, how regulatory oversight can remain effective as AI scales and adapts in real-world settings, and how patient-centered innovation can be operationalized as a concrete governance standard rather than an aspirational principle. The observations and recommendations that follow are intended to support HHS in advancing governance approaches that are scientifically grounded, practically implementable, and aligned with the realities of AI deployment in clinical and public-sector systems.

AI is no longer experimental within health care. It is embedded within clinical decision support, utilization management, documentation, triage, and surveillance systems. As a result, the dominant risks associated with AI are no longer confined to algorithm design or pre-deployment performance. They emerge after deployment, through interaction with clinicians, workflows, institutions, and populations. This shift has profound implications for how AI should be evaluated and governed.

At present, clinical AI is largely regulated as a discrete product. In practice, it functions as a population-level intervention.

Once introduced into a health system, an AI system does not simply generate outputs. It reshapes how decisions are made, how attention is allocated, how workflows evolve, and how care is delivered across groups of patients. These effects accumulate over time and extend beyond the individuals directly targeted by the algorithm. From a regulatory standpoint, this distinction is critical, because the most consequential harms and benefits of AI are not immediate or isolated. They are systemic.

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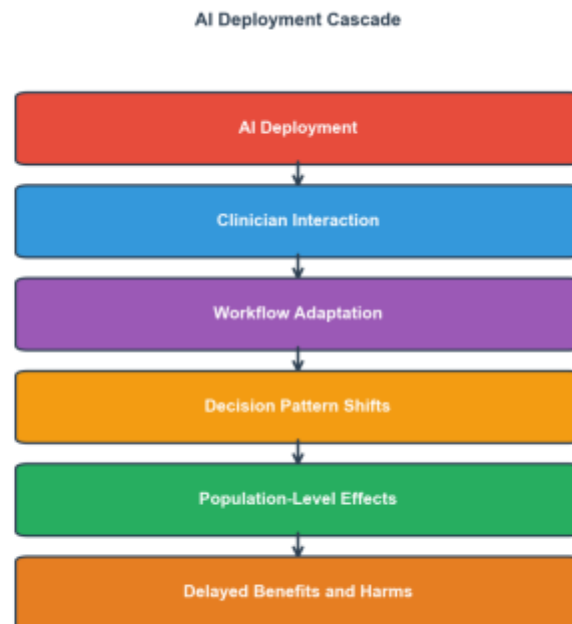
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The operational reality of deployed AI can be represented as a causal sequence rather than a static endpoint:

Current regulatory frameworks largely evaluate AI at the point of deployment, near the top of this sequence. However, the majority of risk emerges downstream, where no systematic oversight currently exists.

Pre-deployment evaluation, while necessary, is structurally insufficient. Validation studies answer whether an algorithm can perform under specified conditions, but they do not answer



whether the system remains safe, effective, or equitable once it reshapes real-world clinical behavior. Questions such as who is most frequently exposed to AI-mediated decisions, how reliance on AI changes clinician judgment, how performance drifts over time, or how outcomes differ across subpopulations cannot be resolved before deployment. They require observation in practice.

This limitation is not theoretical. In one evaluation I conducted, an AI model demonstrated strong pre-deployment performance across conventional metrics. After implementation, clinicians increasingly relied on the model for triage decisions. High-risk patients were identified more efficiently, but moderate-risk patients experienced longer delays. No alerts were triggered. No technical failures were logged. The model behaved exactly as designed. The harm emerged through interaction with workflow and human adaptation, not through algorithmic error.

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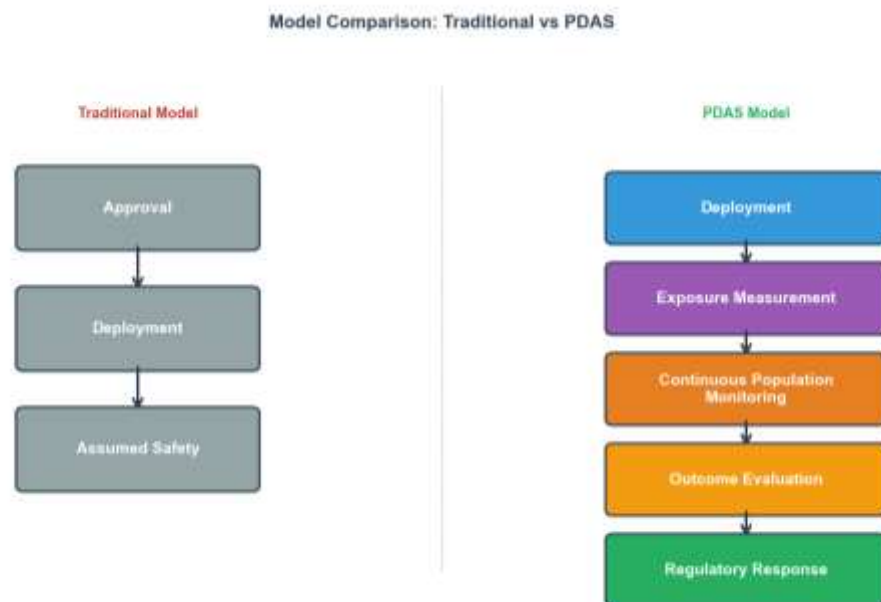
Because no post-deployment monitoring was required, the effect remained invisible until outcome disparities became apparent months later.

From an epidemiologic perspective, this pattern is familiar. Deployed AI exhibits properties that closely mirror population-level exposures. It has intensity, reflecting how frequently it influences decisions. It has latency, with effects that manifest downstream. It demonstrates effect modification, producing different impacts across demographic, clinical, and social groups. It generates spillover effects that influence patients and clinicians not directly targeted. Its risk profile changes over time as data, workflows, and institutional practices evolve.

Yet unlike other population-level interventions, AI is not subject to routine post-deployment surveillance.

This gap represents a regulatory science failure rather than a technological one. HHS has long recognized that interventions with diffuse, delayed, or heterogeneous effects cannot be governed through one-time approval alone. Pharmacovigilance, device post-market surveillance, and public health monitoring systems all exist because pre-market evidence is insufficient to ensure safety at scale. Clinical AI currently sits outside this paradigm.

To address this gap, I recommend that HHS adopt **Post-Deployment Algorithmic Surveillance (PDAS)** as a cross-agency framework for governing clinical AI. PDAS reflects a simple principle: **no AI system influencing care should be presumed safe or effective based solely on pre-deployment evidence.**



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Under PDAS, evaluation shifts from a static, product-based model to a dynamic, population-based model. This shift can be illustrated as follows:

The first element of PDAS is exposure characterization. Without knowing who is exposed to AI-mediated decisions, how often, and in what contexts, no causal assessment is possible. Exposure measurement is foundational, yet it is rarely required.

The second element is continuous population-level performance monitoring. AI systems must be evaluated longitudinally, not episodically. Calibration, discrimination, and error patterns should be tracked over time, with mandatory stratification by key subgroups. Performance drift should be treated as a safety signal, not a technical footnote. In another deployment I reviewed, a model's performance gradually degraded due to changes in coding practices within the electronic health record. Because no routine monitoring was in place, disparities widened quietly over a year. The model did not fail. Oversight did.

The third element is outcome-linked evaluation. After deployment, accuracy alone is not an adequate metric of success. The relevant question becomes whether AI use improves downstream clinical outcomes, access, or utilization patterns. Surrogate metrics that are acceptable during development become insufficient once systems influence care at scale.

The fourth element is structured reporting of algorithmic harms. Many AI-related harms do not resemble traditional adverse events. They include delayed care, silent misclassification, workflow distortions, and disparity amplification. These events rarely trigger complaints or error reports, yet they accumulate into population-level harm. PDAS requires mechanisms to capture these signals systematically.

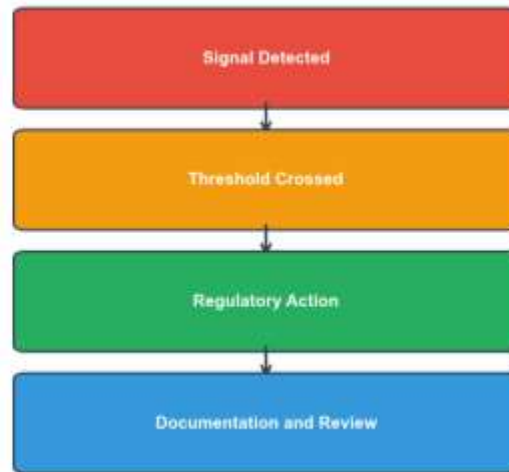
The fifth element is predefined regulatory response pathways. Surveillance without action is performative. PDAS requires clarity regarding when recalibration, restriction, suspension, or withdrawal is warranted, and how those actions are documented and enforced.

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Signal Detection and Response



Patient-centered innovation must be understood within this same framework. Patient-centeredness is often treated as an ethical aspiration or design preference. From a safety perspective, this framing is inadequate. An AI system that influences care but cannot be explained, contested, or reviewed is not merely non-patient-centered; it is unsafe. I have encountered cases in which AI-driven utilization decisions delayed care, clinicians could not explain the rationale, patients had no appeal pathway, and accountability was diffuse. Technical accuracy was irrelevant. The failure was one of governance. Patient contestability and human accountability must be treated as regulatory requirements, not optional features.

Equity considerations further reinforce the need for post-deployment surveillance. Disparities associated with AI are frequently attributed to biased training data, but many inequities emerge only after implementation. Uneven adoption, differential trust, resource constraints, and context-specific failures all contribute to disparate impact over time. Equity cannot be ensured through design alone. It must be measured continuously. PDAS should therefore include equity-specific monitoring requirements for AI deployed in Medicaid, maternal health, rural care, and public health settings.

Adaptive and continuously learning AI systems introduce additional complexity. Static approval models are incoherent when systems are designed to evolve. Regulation must shift toward process-based accountability, with predefined learning boundaries, rollback mechanisms, transparent change logs, and independent auditability. Without these safeguards, adaptation becomes experimentation embedded within routine care.

Finally, none of this is feasible without interoperability. Surveillance, auditability, contestability, and equity monitoring all depend on the ability to access and integrate data across systems.

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Vendor opacity and lock-in are not merely market concerns. They are governance failures. HHS's existing interoperability initiatives provide a necessary foundation, but AI oversight must be explicitly built upon them.

In closing, artificial intelligence will continue to reshape health care regardless of regulatory posture; the question before **U.S. Department of Health and Human Services** is whether that transformation is governed with the same rigor applied to other population-level interventions. Population science already provides the tools needed to meet this challenge. Post-Deployment Algorithmic Surveillance (PDAS) advances the Department's stated goals by establishing a practical framework for continuous evaluation, accountable governance, and patient-centered oversight of AI systems once they are embedded in care. By shifting evaluation from static performance metrics to population-level outcomes, enabling timely regulatory response to emerging risks, and ensuring that patients remain visible and protected within AI-mediated systems, PDAS allows innovation to proceed without normalizing unmeasured harm. In this way, PDAS supports the responsible adoption of AI while maintaining public trust, safeguarding equity, and strengthening the evidence base for AI in health and public health.

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Sincerely,



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